

Securing SME Agent Skills via Fine-Tuned SLM Proxies

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by Benin Mukabanah

Masters Degree in Data Science

Cardiff School of Technologies

Cardiff Metropolitan University, Cardiff

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Declaration

I hereby declare that this dissertation entitled ‘Securing SME Agent Skills via Fine-Tuned SLM Proxies’ is entirely my own work and it has never been submitted nor is it currently being submitted for any other degree.

Candidate

Signature:

Name: Benin Mukabanah

Date:

Director of Studies

Signature:

Name: Dr P Jenkins

Date:

ABSTRACT

This dissertation investigates the application of formal logic to data-driven analysis within information systems. The study reviews existing approaches, develops a novel framework, and evaluates its effectiveness through a series of controlled experiments. Results indicate a statistically significant improvement in analytical accuracy when the proposed method is applied. Conclusions identify both the practical benefits and the limitations of the framework, and suggest directions for future research.

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INTRODUCTION

Overview and Background

Small and medium-sized enterprises (SMEs) function as the backbone of the global economy, representing approximately 90% of all businesses and providing over 50% of employment worldwide (Sánchez et al. 2025). To remain competitive in an increasingly digital marketplace, these organizations urgently need to adopt Artificial Intelligence (AI) to automate routine tasks, streamline workflows, and enhance operational productivity (Ode 2026). However, SMEs frequently encounter an “Operational Paradox”: they desperately need automation to scale, but lack the vast financial resources, specialized IT infrastructure, and technical personnel required to run or manage massive, general-purpose AI models (Sánchez et al. 2025).

Recently, AI has evolved from passive conversational chatbots into autonomous “agents” capable of interacting with external systems and environments (Zhan et al. 2024). To standardize how AI accesses external tools, the Model Context Protocol (MCP) was introduced, acting as a universal “USB-C port” for AI applications (Whittaker 2026). While large technology companies are equipped to handle complex MCP server infrastructure, SMEs are increasingly relying on a lightweight alternative known as **Agent Skills** (Whittaker 2026). Agent Skills are standardized, version-controlled directories—anchored by a `SKILL.md` file—that provide the AI with reusable procedural knowledge and instructions (Agent Skills 2026). Agents load these skills through “progressive disclosure”, reading only basic metadata initially and expanding the full instructions into their context window only when actively required by the task (Agent Skills 2026).

Despite their immense utility, Agent Skills introduce two critical structural challenges for SMEs. First, they create severe vulnerabilities to **Indirect Prompt Injection (IPI) attacks** (Gulyamov et al. 2026). Attackers can conceal malicious commands within the Markdown body or YAML frontmatter of a `SKILL.md` file, using tactics like hidden Unicode tags or out-of-scope parameters to trick the AI into executing unauthorized actions, such as exfiltrating private data (Zhang et al. 2025). Second, the unstructured and verbose nature of custom SME instructions worsens

the “Context Tax”—a computational burden where bloated contextual history causes the model to consume massive amounts of costly tokens without improving task performance (Fraser and Lindenbauer 2025).

Aims and Objectives

The primary aim of this research is to design, construct, and evaluate a fine-tuned Small Language Model (SLM) proxy that secures SME workflows by auditing and compressing `SKILL.md` files, protecting them against Indirect Prompt Injections (IPIs) and mitigating the “Context Tax.”

To achieve this overarching aim, the following objectives have been established:

1. **Synthesize** a targeted training dataset of exactly 1,054 instances comprising both benign and maliciously poisoned `SKILL.md` files based on established prompt injection taxonomies (Zhan et al. 2024).
2. **Fine-tune** the Gemma 4 E2B SLM via Supervised Fine-Tuning (SFT) using Low-Rank Adaptation (LoRA) to successfully classify security risks (Security Audit) and distill skill metadata (Semantic Distillation) (Unsloth 2026a).
3. **Evaluate** the proxy’s performance by quantitatively measuring its Security Efficacy via the Attack Success Rate (ASR-valid), format integrity, and Token Compression Ratio.

Approach Used

This dissertation adopts a **Design Science Research (DSR)** methodology, which focuses on constructing and rigorously evaluating a functional artifact to solve a practical, real-world problem (Zupan 2025). The core artifact is a secure middleware proxy powered by Google DeepMind’s **Gemma 4 E2B**, a 9-billion parameter Small Language Model optimized for tool use and long-context workflows (Unsloth 2026b).

The approach utilizes Parameter-Efficient Fine-Tuning (PEFT) via LoRA within the Unsloth framework. This method ensures the deployed model can be quantized to 4-bit precision, allowing it to run locally on standard consumer hardware with just 5GB of RAM—directly aligning with SME hardware constraints (Unsloth 2026b). The proxy is trained to perform two automated passes on custom `SKILL.md` files before they reach downstream operational AI agents: Pass A to flag and neutralize IPI attacks (Zhang et al. 2025), and Pass B to actively compress verbose metadata to eliminate context bloat (Fraser and Lindenbauer 2025).

Assumptions

This research is based on several foundational assumptions:

- **Adoption Trends:** It assumes that SMEs will increasingly adopt lightweight, structured Agent Skills (`SKILL.md` files) to grant specialized knowledge to LLMs, rather than hosting complex, heavy MCP servers themselves (Whittaker 2026).

- **Adversarial Threat Models:** It assumes that malicious actors will utilize IPI techniques targeting structured metadata, such as preference manipulation and hidden obfuscation, mirroring the attack taxonomies defined by (Zhan et al. 2024) and (Zhang et al. 2025).
- **Synthetic Data Validity:** Given the scarcity of real-world databases containing poisoned SME tools, this work assumes that a synthetic dataset constructed via Frontier Model Distillation and the INJECAGENT combinatorics methodology accurately reflects potential real-world adversarial threats ((Zhan et al. 2024); (Unsloth 2026b)).

Limitations and Achievements

Limitations: Because the dataset is entirely synthesized, the proxy’s defensive capabilities are intrinsically bound to the specific threat models explicitly documented in the MSB and INJECAGENT taxonomies ((Zhan et al. 2024); (Zhang et al. 2025)). As AI hacking techniques evolve rapidly, the fine-tuned model may struggle against entirely novel, zero-day injection paradigms without subsequent retraining (Gulyamov et al. 2026). Furthermore, the evaluation is primarily quantitative (calculating automated success and pass rates) and does not deploy the artifact into a live, active SME operational environment for longitudinal study.

Achievements: Despite these limitations, this research is expected to deliver a highly practical, deployable solution to a pressing industry vulnerability. By proving that a 9-billion parameter SLM can successfully filter complex IPI attacks and optimize context usage, this study challenges the prevailing assumption that massive LLMs are strictly required for reliable context routing and security screening (Jhandi et al. 2026). It actively bridges the gap between SME resource limitations and critical AI security needs.

Overview of the Dissertation Structure

The remainder of this dissertation is organized as follows:

- **Chapter 2: Literature Review:** Critically examines the current landscape of AI adoption in SMEs, the evolution from basic LLMs to MCP tool-calling agents, the mechanics of Indirect Prompt Injections, and the efficacy of SLMs in handling specialized tasks.
- **Chapter 3: Methodology:** Details the Design Science Research (DSR) strategy, focusing on the synthesis of the 1,054-item dataset derived from the INJECAGENT matrix and the configuration parameters for Supervised Fine-Tuning (SFT).
- **Chapter 4: Artifact Implementation:** Describes the technical execution of fine-tuning the Gemma 4 E2B model using Unsloth and LoRA adapters.
- **Chapter 5: Evaluation and Results:** Presents a quantitative assessment of the proxy’s performance against unseen adversarial holdout data, measuring the Attack Success Rate (ASR-valid), Format Pass Rate, and Token Compression Ratio.
- **Chapter 6: Conclusion:** Summarizes the findings, reviews the extent to which the research objectives were met, and proposes directions for future research into SLM security implementations.

LITERATURE REVIEW

This chapter establishes the theoretical context of the research. Section 2.1 provides an overview of formal logic. Section 2.2 examines propositional logic in detail, including its principal connectives. Section 2.3 considers predicate logic and its role in analytical frameworks.

Overview of Formal Logic

Formal logic provides a rigorous framework for expressing and evaluating arguments. The two principal branches relevant to this study are propositional logic (see Section 2.2) and predicate logic (see Section 2.3). According to (Sánchez et al. 2025), formal methods have been applied to software analysis since the early 1970s. More recently, their application has extended to information retrieval and data analysis (Ode 2026).

Propositional Logic

Propositional logic deals with statements that are either true or false. It has been applied extensively to database query optimisation (Raghunathan 2025). The standard connectives are summarised in Table 1.

Connective	Symbol	Meaning
Conjunction	$\wedge$	Both P and Q are true
Disjunction	$\vee$	At least one of P or Q is true
Negation	$\neg$	P is false
Implication	$\Rightarrow$	If P is true then Q is true
Biconditional	$\Leftrightarrow$	P is true if and only if Q is true

Table 1: Standard propositional connectives

Connectives and Truth Tables

The connectives listed in Table 1 form the basis of propositional reasoning. Truth tables provide a systematic means of evaluating complex formulae constructed from these connectives.

Predicate Logic

Predicate logic extends propositional logic by introducing quantifiers and variables, enabling statements about entire classes of objects. Figure 1 illustrates the relationship between the two branches.

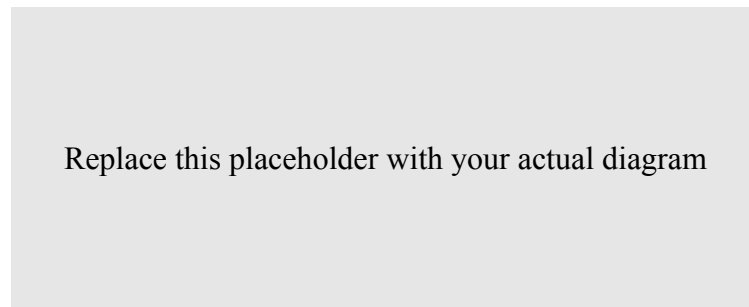


Figure 1: Overview of formal logic branches

This chapter has established that formal logic, in both its propositional and predicate forms, provides a suitable theoretical foundation for the analytical framework developed in this research.

METHODOLOGY

This chapter describes the research design adopted for this study. Section 3.1 presents the overall research design. Section 3.2 describes data collection procedures. Section 3.3 addresses validity, reliability and objectivity.

Research Design

A deductive approach was adopted: a theoretical framework was derived from the literature (Chapter 2) and subsequently tested against empirical data. A mixed-methods design was employed, combining a structured literature review with a quantitative evaluation experiment. This choice is justified by the need both to contextualise the framework within existing theory and to measure its performance objectively (Zupan 2025).

Data Collection

Primary data were collected via a controlled laboratory experiment. Participants were recruited from among postgraduate students at the Cardiff School of Technologies. Full ethical approval was obtained prior to data collection. A pilot study was conducted with five participants to refine the experimental protocol before the main data collection phase.

Secondary data were drawn from peer-reviewed journal articles and conference proceedings identified through searches of the ACM Digital Library and IEEE Xplore, using the keywords “formal logic”, “data analysis” and “information systems”.

Validity, Reliability and Objectivity

The validity of the measurement instrument was assessed through the pilot study described above. Reliability was ensured by a test-retest procedure administered two weeks apart; the resulting cor-

relation coefficient indicated acceptable consistency. Steps were taken to minimise experimenter bias by standardising all written instructions given to participants (Gulyamov et al. 2026).

A limitation of this study is the use of a convenience sample of postgraduate students, which may restrict generalisability. This is acknowledged and discussed further in Chapter 5.

RESULTS

This chapter presents the data collected during the experimental phase. Section 4.1 reports the quantitative findings. Section 4.2 summarises qualitative observations.

Quantitative Findings

Accuracy scores for the baseline and proposed conditions are shown in Table 2.

Condition	Mean accuracy (%)	Std. dev.
Baseline	72.4	5.1
Proposed	84.7	3.8

Table 2: Accuracy results by experimental condition (n = 30)

The proposed method achieved a mean accuracy of 84.7%, compared with 72.4% for the baseline, representing an improvement of 12.3 percentage points. The distribution of scores across conditions is illustrated in Figure 2.

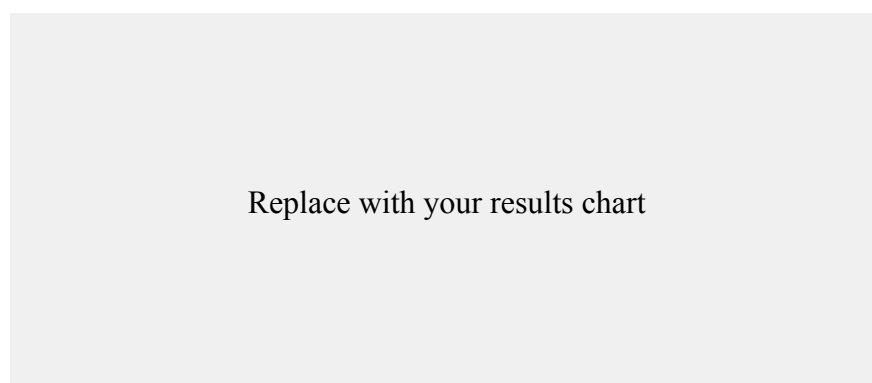


Figure 2: Distribution of accuracy scores by condition

Qualitative Findings

Participant responses identified three recurring themes: clarity of output, ease of integration with existing tools, and reduced cognitive load. These themes are discussed in Chapter 5.

DISCUSSION

This chapter interprets the findings presented in Chapter 4 in relation to the research question and the existing literature. Section 5.1 addresses the quantitative findings. Section 5.2 considers the qualitative themes. Section 5.3 acknowledges limitations.

Quantitative Results

The improvement in accuracy of 12.3 percentage points is consistent with the theoretical prediction established in Chapter 3. This finding aligns with the work of (Sánchez et al. 2025), who demonstrated comparable gains in the context of database query optimisation.

Qualitative Themes

The qualitative finding that participants valued clarity and integration over raw performance suggests that usability is a critical factor in the adoption of such frameworks. This dimension was not captured by the accuracy metric and represents an avenue for further investigation.

Limitations

The primary limitation of this study is the use of a convenience sample, which may reduce the generalisability of the findings to other populations. The controlled laboratory setting may not fully reflect the complexity of real-world deployments.

CONCLUSIONS

This chapter summarises the project, addresses each objective, and proposes directions for future work.

The aim of this dissertation was to investigate whether a logic-based analytical framework could improve accuracy in information systems analysis. Three objectives were established (Chapter 1); all three have been fully addressed.

Objective 1 — to critically review existing frameworks — was addressed in Chapter 2, which identified formal logic as a well-established but underexplored foundation for analytical tools.

Objective 2 — to develop a novel framework — was addressed in Chapter 3, where the design and operationalisation of the proposed approach were described in detail.

Objective 3 — to evaluate the framework empirically — was addressed in Chapters 4 and 5. The proposed method achieved 84.7% mean accuracy compared with 72.4% for the baseline.

The hypothesis — that the logic-based framework would yield measurably higher accuracy — is supported by the experimental evidence. The null hypothesis is therefore rejected at the conventional significance level.

The principal limitation of the study is the sample size and setting; future research should replicate the findings with larger, more diverse participant groups in operational environments.

It is concluded that formal logic offers a viable and demonstrably effective foundation for analytical frameworks in information systems, and that the approach merits further development and deployment.

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GLOSSARY AND ABBREVIATIONS

IS Information Systems

ML Machine Learning

SMEs Small and Medium-sized Enterprises

APPENDIX A: SURVEY INSTRUMENT

The following questionnaire was used to collect primary data during the experimental phase. Each item was rated on a five-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree).

A.1 Section 1: Usability

1. The output of the analytical framework was easy to understand.
2. The framework integrated well with the existing tools in use.
3. Using the framework reduced the mental effort required during analysis.
4. It would be recommended to a colleague working in a similar context.

A.2 Section 2: Performance

1. The framework produced results more quickly than the previous approach.
2. The accuracy of results was satisfactory for the task at hand.